

Solutions – Practice Problem Set 3

ECON 320 – Introduction to Mathematical Economics

Fall 2025

1. Nonlinear Production Optimization

$$Q(L) = 50L^{0.6}e^{-0.02L}$$

Solution Steps

1. **Derivative:**

$$\frac{dQ}{dL} = 50e^{-0.02L} (0.6L^{-0.4} - 0.02L^{0.6}) = 50e^{-0.02L} L^{-0.4} (0.6 - 0.02L)$$

2. **FOC:** $0.6 - 0.02L = 0 \Rightarrow L^* = 30$.

3. **SOC:** The term $(0.6 - 0.02L)$ switches from positive to negative, implying $Q''(30) < 0$.
Hence L^* gives a **maximum**.

4. **Interpretation:** At $L < 30$, increasing labor raises output; beyond $L = 30$, diminishing returns dominate.

Key Concept: The product of a power and an exponential term often produces an interior maximum because growth from the power term is eventually offset by exponential decay.

2. Utility with Exponential Preferences

$$U(x) = 5x^{0.4}e^{-0.1x}$$

Steps

$$\begin{aligned}U'(x) &= 5e^{-0.1x}(0.4x^{-0.6} - 0.1x^{0.4}) \\ &= 5e^{-0.1x}x^{-0.6}(0.4 - 0.1x)\end{aligned}$$

$$0.4 - 0.1x = 0 \Rightarrow x^* = 4$$

$$U''(x) = 5e^{-0.1x}[-0.24x^{-1.6} - 0.04x^{-0.6} + 0.01x^{0.4}]$$

Since $U''(4) < 0$, a maximum exists.

$$U(4) = 5(4^{0.4})e^{-0.4} \approx 5.84$$

Economic Meaning: Utility rises then falls—capturing *satiation*. The exponential term reflects diminishing satisfaction beyond moderate consumption.

3. Logarithmic Profit Maximization

$$\pi(Q) = 200 \ln(Q + 1) - 30Q$$

Steps

$$\pi'(Q) = \frac{200}{Q + 1} - 30 = 0 \Rightarrow Q^* = \frac{200}{30} - 1 = 5.67$$

$$\pi''(Q) = -\frac{200}{(Q + 1)^2} < 0 \Rightarrow \text{Maximum.}$$

$$\pi(5.67) = 200 \ln(6.67) - 170.1 \approx 209.3$$

Interpretation: The logarithmic form of revenue implies strongly diminishing marginal profit; each additional unit contributes less due to the concavity of $\ln(Q + 1)$.

4. Advertising Response with Diminishing Returns

$$S(A) = 400(1 - e^{-0.05A}) - 2A$$

Derivative:

$$S'(A) = 20e^{-0.05A} - 2 = 0 \Rightarrow e^{-0.05A} = 0.1 \Rightarrow A^* = 46.05$$

$$S''(A) = -1e^{-0.05A} < 0 \Rightarrow \text{Maximum.}$$

$$S(46.05) = 400(1 - 0.1) - 92.1 = 267.9$$

Interpretation: The exponential term ensures sales approach an upper limit (saturation). Advertising is most effective at low levels and yields diminishing returns as A grows.

5. Elasticity via Logarithmic Differentiation

$$C(Q) = 10Q^{0.7}(2 + \ln Q)$$

Take logs:

$$\ln C = \ln 10 + 0.7 \ln Q + \ln(2 + \ln Q)$$

Differentiate:

$$\begin{aligned} \frac{1}{C} \frac{dC}{dQ} &= \frac{0.7}{Q} + \frac{1}{Q(2 + \ln Q)} \\ E_{CQ} &= \frac{dC}{dQ} \frac{Q}{C} = 0.7 + \frac{1}{2 + \ln Q} \end{aligned}$$

At $Q = 10$:

$$E_{CQ} = 0.7 + \frac{1}{2 + 2.3026} = 0.917$$

Interpretation: The elasticity is less than one—costs rise less than proportionally to output, showing economies of scale.

6. Maximization with Rational Function

$$R(P) = \frac{100P}{4 + P^2}$$

$$R'(P) = \frac{100(4 - P^2)}{(4 + P^2)^2} = 0 \Rightarrow P^* = 2$$

$$R''(P) < 0 \text{ at } P = 2 \Rightarrow \text{maximum.}$$

$$R(2) = \frac{100(2)}{8} = 25$$

Interpretation: At $P^* = 2$, revenue peaks as further price increases reduce quantity enough to offset higher price.

7. Cost Minimization with Logarithmic Term

$$C(x) = 20x - 50 \ln x$$

$$C'(x) = 20 - \frac{50}{x} = 0 \Rightarrow x^* = 2.5$$

$$C''(x) = \frac{50}{x^2} > 0 \Rightarrow \text{minimum.}$$

Interpretation: The $-\ln x$ term captures scale economies—average cost decreases initially then increases linearly beyond x^* .

8. Growth Model with Continuous Time

$$P(t) = 800e^{0.03t} - 400e^{-0.02t}$$

$$P'(t) = 24e^{0.03t} + 8e^{-0.02t}$$

$$P''(t) = 0.72e^{0.03t} - 0.16e^{-0.02t} = 0 \Rightarrow e^{0.05t} = 0.222 \Rightarrow t^* = -30.1$$

Growth rate highest initially; declines over time.

Interpretation: The first exponential drives long-run expansion, while the second captures early-stage decay or adjustment.

9. Logarithmic Demand and Revenue

$$P(Q) = 60 - 5 \ln(Q + 1), \quad R(Q) = Q(60 - 5 \ln(Q + 1))$$

$$R'(Q) = 60 - 5 \ln(Q + 1) - \frac{5Q}{Q + 1} = 0$$

Numerical testing shows revenue rises gradually, approaching an upper limit. Logarithmic demand implies continually decreasing sensitivity; revenue increases slowly but without a sharp interior maximum.

10. Challenge: Nonlinear Utility and Elasticity

$$U(x) = \ln(3x + 2) - 0.5x^2$$

$$U'(x) = \frac{3}{3x + 2} - x = 0 \Rightarrow 3 = 3x^2 + 2x \Rightarrow x^* = 0.721$$

$$U''(x) = -\frac{9}{(3x + 2)^2} - 1 < 0 \Rightarrow \text{maximum.}$$

$$U(0.721) = \ln(4.163) - 0.5(0.721)^2 = 1.426 - 0.26 = 1.166$$

Interpretation: Utility rises with consumption initially but declines for large x as the quadratic term dominates, capturing disutility of overconsumption.